

Privacy-Aware Respiratory Symptom Detection in-the-wild

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ABSTRACT

Coughs and sneezes are important symptoms of respiratory, allergic and Influenza-Like Illnesses. Monitoring the number of occurrences of these symptoms can not only track patients' health but also help with understanding the dynamics of outbreaks. While there has been extensive research around detecting coughs, creating reliable methods to identify sneezes are still unexplored, partially owing to scarce availability of data. Most respiratory symptom detection algorithms operate at high sampling rates and thereby are privacy-invasive. By downsampling the data extensively, we can ensure that the speech cannot be reconstructed, yet retain enough signal characteristics to identify sneeze and cough sounds. Using a lightweight machine learning based approach, we propose a privacy-aware and energy-efficient pipeline that can detect coughs and sneezes. Such a system can lead to higher adoption and thus make tracking diseases in individuals and communities easier.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

KEYWORDS

Acoustic Activity Detection, Respiratory Symptoms, Privacy-Aware Sensing

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1 INTRODUCTION

Respiratory illnesses are generally characterized by coughs and sneezes. Estimating the frequency and counts of these symptoms are strong indicators of illness progression and severeness [8, 12]. Sneezes are also highly associated with allergies and can be used to track severity [13, 20]. Finally, detecting these signals can lead to timely diagnosis of chronic diseases such as COPD/Asthma [17].



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Hence automated detection of coughs and sneezes can help us monitor the health of such individuals better and even ensure timely escalation if required. These symptoms and their frequencies can also be used for modeling common cold, Influenza-Like Illnesses (ILI) [19] and very recently COVID-19 [16]. At a community level, tracking such activities can also help us model the outbreaks of ILI, a CDC top-10 infectious disease¹.

1.1 Respiratory Symptom Detection

There has been extensive research on detecting cough activity. Prior research has explored how to create cough detection systems using audio [3] and motion-related signals [22]. In contrast, there has been limited progress toward identifying sneezing activity, leaving a huge gap toward *Sneezing Activity Detection* for humans. The closest to identifying cough and sneeze sounds that use a lightweight approach is [6], which uses MFCC as the features for a time-warping based classifier. MFCCs are known to be robust descriptors of speech even at low sampling rates [11] and hence the speech characteristics might often be retained, thereby compromising on privacy.

1.2 Lightweight Privacy-Aware Algorithms

Due to the benefits explained above, there is a strong need for developing an activity detection that can identify sneezing in addition to coughing in individuals. Recent advances in acoustic modeling of respiratory diseases and outbreaks use audio signals at a high sampling rate. This makes them privacy-invasive since the audio data can contain sensitive information that can be easily leaked to the wrong hands. Speech Obfuscation [14] and eigenvalue-based transformations [11] have shown promising results in solving this problem, but the system operates at a high sampling rate, and thus it is energy hungry. Recent approaches in cough detection have shown that downsampling the signals to a large extent makes reasonable reconstruction nearly impossible [1, 6].

We use a similar approach to show that we can extract sneezing and coughing sounds while being agnostic to speech sounds. Since we operate at lower sampling rates, our system is comparably energy-efficient. Further, since our model is trained and data sampled from various sources, it can perform well in also in-the-wild. Privacy here is defined in terms of the perceived level of degradation of speech [1, 11]. Popular measures of perceived quality are PESQ [18] and VisQOL [10].

Our Contributions

The objective of our work is to enable continuous disease progression monitoring by allowing individuals to be continuously

¹<https://www.cdc.gov/nchs/fastats/infectious-immune.htm>

monitored and promptly alert their healthcare providers. This necessitates the development of audio applications that can operate without access to speech while being lightweight for continuous usage. As established above, cough and sneezing detection for humans from audio sampling can provide essential contributions to public health, but existing audio-based algorithms are highly invasive and energy-hungry.

To address these limitations, we propose a new pipeline to support this use case, as shown in Figure 1. In summary, we design a joint symptom tracker that can identify sneeze and cough activity in-the-wild. Even at frequencies as low as 1kHz, our pipelines can detect these symptoms at an overall 70% and 57% sensitivity. We also then evaluate the model based on its features and present a high level discussion of the salient aspects of our pipeline.

2 DATASET

While coughs have been widely studied, there is little data about other sickness-related sounds. The Google AudioSet repository [9] contains videos from YouTube that were manually annotated for their content. FluSense [2] generously released their annotations for [9] that contain various sounds such as cough, sneeze, sniffle, and throat-clearing.

The data contains 50,920 seconds of audio sounds collected from various sources in the wild. Since the data is sourced from many different locations and situations, there is already an extensive amount of noise in it. Further, as shown in Table 1, we see that the classes of interest, and particularly cough and sneeze are very rare. If a model can extract respiratory symptoms from this data, then we can claim that the model may perform better than chance also in a real-world setting.

	Duration (in seconds)	Number of instances
Cough	7662.17	2486
Sneeze	654.71	611
Other	42602.79	8591
Total	50919.67	11688

Table 1: Distribution of the dataset across the three classes

We use this data to build classifiers that can detect cough, sneeze, and other activities. To ensure that the model is robust, we keep other human sounds and silences in other sections. Such distribution of labels can help us distinguish all human sounds from the signals of interest.

3 PIPELINE

The audio signals are passed through different feature extractors and the generated features are classified into {cough, sneeze, other}. Figure 1 describes the training flow of our pipeline. This section provides specifics of our pipeline.

3.1 Resampling

Since the data is collected from various sources, we want to first bring data to a uniform level to create a baseline for comparison. We first resample all the audio data to 16 kHz. This will be the highest sampling rate in our experiments. We follow a downsampling approach similar to Abdelkhalek et al. [1] where each audio

signal when converted from f_{old} to f_{new} is first downsampled, then aliased. Using this we generate audio signals for sampling rates $f_s = \{500, 1000, 2000, 4000, 8000\}$ Hz. Since the human ear is sensitive to sounds only above 4kHz [1], we primarily focus on signals below this cutoff.

3.2 Feature Design

Using the audio signals, we derive **23 features** that describe each audio frame. Since cepstral coefficients are often used in speech recognition itself [11], to ensure the models learn non-speech features, we do not use MFCC features despite their proven efficacy in cough detection [2, 6]. Table 2 shows a summary of the extracted features.

Spectral Features

The spectral components describe frequency spectrum based information about the signal. We use the following spectral features for our task:

- *Chroma STFT*: Chromagrams map the entire spectral audio characteristics across octaves. We extract **12 bins** per frame out of the chromagrams to represent the frequency distribution of the frame along these octaves
- *Spectral Contrast*: divides the spectrogram into sub-bands and compares their energy with respect to higher and lower quantiles. Using sub-bands, we generate **four** such features
- *Spectral Rolloff*: measures the frequency range which contains over 85% of the max energy. A higher spread will imply a larger rolloff
- *Spectral Flatness*: measures how close a spectrum is to White Gaussian Noise. We believe this measure can separate frames with noise or silence versus those with sound successfully
- *Spectral Centroid*: tracks the mean of an (assumed) Gaussian distribution of the normalized energy across frequencies
- *Spectral Bandwidth*: measures the p-norm of the spectral distribution across frequencies. We measure bandwidth at $p=2$
- *Band Energy Ratio*: measures the ratio of the energy of sub bands above and below a certain threshold frequency f_{th} . From an exploratory analysis of the spectrum of the classes, we set f_{th} as 500 Hz

Time-Domain Features

In the time domain, for each frame that we analyze, we extract two features.

- *Root Mean Square*: The root mean square of the signal amplitudes gives a measure of the overall signal energy in a given frame
- *Zero Crossing Rate*: Audio signals with a high frequency tend to oscillate faster. Zero crossing rate measures the number of times the signal amplitude changes from negative to positive

3.3 Data Augmentation

After the features are extracted, we only augment the dataset for training. Since the data has an extreme class imbalance in our classes of interest vs others, we apply SMOTE [7], a popular minority oversampling technique. The oversampling step balances the

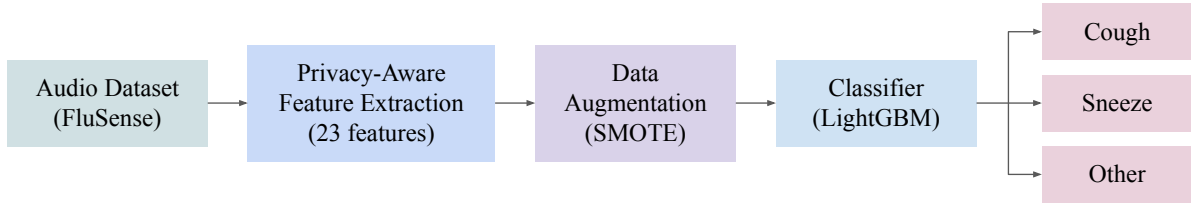


Figure 1: Proposed pipeline for respiratory symptom detection

number of samples in each class. Since the data is already noisy due to different recorders, sources and surroundings, we argue that the dataset might not need further noise augmentation.

3.4 Classifier

Tree based classifiers are largely information driven and hence can provide meaningful explanations for classification. Tree based classifiers have also shown to improve trust in other healthcare settings such as patient-provider interactions [4] and diagnostics [15]. Hence we choose three tree-based classifiers for the classification of the audio features.

Decision Trees: or Binary Trees are the simplest tree-based classifiers. Using information gain or entropy these trees can parse through large datasets and split the dataset into branches until a cluster/leaf cannot be branched further.

Random Forests: use Decision Trees as a base to generate more than one trees (stumps) to create an ensemble of different decision tree predictors, making it much more robust to overfitting.

Gradient Boosted Trees: use gradient as a mechanism to grow the tree with respect to the splits made so far. Since the trees use errors of the previous iterations to make the next iteration, they can capture a higher complexity in the data. We use the LightGBM implementation for its fast and efficient performance.

4 IMPLEMENTATION

The feature resampling and extraction are done using librosa, a python package for audio processing. All the machine learning classifiers are built using scikit-learn, another python package. The Decision Trees were trained with *max_depth* of 5 and with *split_criteria*

as ‘entropy’. Different Random Forest models with varying depth and number of estimators were used. The random state was set to 42 in all cases. All downsampled features were first extracted and stored to reduce the repetition of computation at run-time. Unless specified we keep all the parameters as defaults for reproducibility.

Training Process: We split our data files into 90% and 10% for training and testing. We run a five-fold cross validation on the training set. To ensure no data leakage, all the splits between training, validation and test are performed at the recording (or file) level and not frame/window level. Since we want our trained model to be agnostic to the sampling frequency of the sound, we sequentially train it to learn the same samples across different frequencies. We also optimize the models through various parameter search experiments. From our experiments, we find that the Random Forest model with seven different estimators each with a max depth of 5 and Gradient Boosted Classifier with max depth 4 and seven estimators perform the best overall. The final models are trained upon model selection on the entire training set and tested on the held-out recordings resampled at different sampling frequencies.

Testing Process: To conduct a direct comparison, we ensure that the test data samples from the audio are constant across all the sampling rates.

5 RESULTS

Overall, we demonstrate a better performance using gradient boosted trees compared to alternatives (see Figure 2). This section summarizes our results.

5.1 Model Performance

Figure 2 shows that while applying a sampling frequency that is larger than 1 kHz, the weighted F1-score yielded by all approaches are higher than 0.5. This shows that the features chosen are effective to perform the task. We therefore choose to use LightGBM as our final model. Note that at lower sampling rates, for the same files, we will have lesser samples available for the same hop/frame duration (10/25 ms). Hence the comparison only makes sense within classifiers at a given sampling rate and not with classifiers at different rates.

5.2 Feature Analysis

We plot the feature importance scores or the modified entropy for each feature in the LightGBM model as shown in Figure 3. We find that the Band Energy Ratio (Hi/L0) is the most decisive feature. However as frequency changes, from Figure 2, we observe that the model performance first increases then drops. We estimate that

Domain	Feature	Number
Spectral	Chroma STFT	12
	Spectral Contrast	4
	Spectral Rolloff	1
	Spectral Flatness	1
	Spectral Centroid	1
	Spectral Bandwidth	1
	Band Energy Ratio	1
	Zero Crossing Rate	1
Time	Root Mean Square	1
Total	Feature Set	23

Table 2: Chosen features across different domains



Figure 2: Performance at different sampling rates in Hz. Our models outperform baseline for all frequencies beyond 1 kHz

since the band features were designed with a cutoff frequency of 500 Hz, the Nyquist sampling limit kicks in and the frequency concentrations in the High and Lo band start becoming indistinguishable. When trained independently for each sampling frequency, we also note that for lower sampling rates (below 4000 Hz), the model lays higher importance to spectral flatness than the band energy ratio.

5.3 Measures of privacy

The tool (VISQOL) [10] is used to assess speech reconstructions through perceived quality. It compares a signal to its degraded version, generating a rating of perceived speech quality from 1 to 5. Studies [1, 11] have shown that downsampled speech signals cannot be upsampled to reconstruct intelligible speech. Abdelkhalek et al. [1] demonstrated a 68% drop in VisQOL ratings when sampling cough/speech sounds at lower frequencies (e.g., 750 Hz instead of 16 kHz), rendering the speech incomprehensible. We used a similar approach on the same dataset as they did and hence expect to observe similar degradation when sampling sound at 16 kHz versus 1 kHz. This significant difference ensures unintelligible speech content, making the system privacy-aware and suitable for secure implementation on edge devices or on-device settings. While this is an inferred measure of privacy, in the future we aim to evaluate comprehensively privacy using measures like PESQ [18, 21], POLQA [5] and qualitative inferences.

5.4 Performance

From repeatedly running predictions at different sampling rates, we see that the time taken in feature extraction is the critical step while all predictors follow log-linear time in prediction. Since all the features are derived based on the raw signal or its spectral distribution, there is no single feature that requires any extra intermediate computational step. Since we resample the audio to lower sampling rates, our total number of frames per second scales down exponentially. Thus the system becomes energy efficient compared to other symptom detectors. We envision this pipeline to be continuously running in the background in ubiquitous devices, primarily wearables. Hence an ideal pipeline should also have low memory consumption at run-time. From a coarse evaluation we

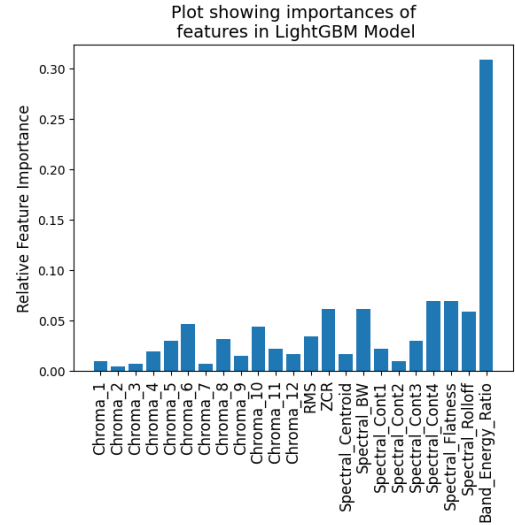


Figure 3: Importance of features in LightGBM Model. We find that Band Energy Ratio is the most decisive feature

find our pipeline (6 MB) to be much smaller compared to deep learning models (45 MB) [1]. However, in the future we aim to train cough-based downsampling pipelines [1, 6] specific for our task and compare their performance as a joint system detectors based on accuracy, memory and latency to establish efficacy of deployment on wearables.

6 CONCLUSION AND FUTURE WORK

This work acts as a start towards sneeze activity detection in a privacy-aware manner. We motivate the need for continuous monitoring and the need for privacy. Inspired from recent trends in cough detection, we create an activity detector that can operate at subsampled (1 kHz) sampling rates while still ensuring promising accuracy. We create spectral features for a one-size-fits-all model to optimize the feature calculation step. By subsampling audio, we also ensure that the pipeline is privacy-aware and comparatively efficient. Due to these characteristics, the system can even be plugged onto wearables and other low power surveillance devices.

More importantly this system acts as a start towards sneeze activity detection in a privacy preserving manner. In the future, we aim to improve the performance by creating more privacy-aware sampling features, more data around less popular respiratory symptoms as well as more powerful interpretable machine learning approaches. We eventually plan to run a user study and collect real-world data in the field to understand our algorithm’s performance better. Collectively these steps could help us unobtrusively track various respiratory symptoms and the lead to improving the overall societal well being.

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